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Optimization of combustion kinetic models on a feasible set

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Abstract

We introduce a new approach to optimizing combustion models by constraining the optimization not only to parameter uncertainties but also to the uncertainties in experimental data. Doing so assures that the best-fit solution produces a model whose predictions deviate the least from the experimental targets while remaining within their experimental uncertainties. The new method is built using the uncertainty-quantification framework of Data Collaboration, and hence allows one to examine the influence of parameter and experiment uncertainties on the optimal solution, and to identify experimental targets that are most difficult to match as well as model parameter values that are likely to be questionable. The proposed methodology is flexible to formulate a variety of objectives and thus to answer seemingly heuristic questions. The technique is demonstrated employing the GRI-Mech 3.0 kinetic model and 76 experimental targets.
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1. Introduction

Detailed chemical-kinetics models used in combustion are composed of a large number of species and reaction steps, whose thermodynamics, transport, and reaction rates are plagued by uncertainties. Such models are typically built by assembling data from diverse sources, which generally does not lead to the degree of confidence and reliability required for useful practical applications [1].

A systematic methodology based on optimization was introduced by Frenklach et al. [2–4]. In that approach, an initial model is optimized within uncertainties of active model parameters to fit a training set of experimental observations. An outcome of this strategy was GRI-Mech; in its latest release, 3.0 [5], a subset of 102 active kinetic and thermodynamic parameters was adjusted within their respective uncertainties to best-fit 77 experimental targets. The accuracy of the chemical-kinetics model developed in this manner has been demonstrated repeatedly (e.g., [6–8]).

The numerical efficiency of the GRI-Mech approach was achieved through Solution Mapping—development of surrogate statistical models for experimental targets via computer experiments [4,9]. Other methods of kinetic-model optimization were developed in recent years [10–12]. The quest, in the meantime, has advanced to quantification of model uncertainty [2,13–25]. One technique in particular, called Data Collaboration [15–18,26], combines the numerical efficiency of Solution Mapping with rigorous deterministic formulation of the uncertainty-quantification problem.

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Data Collaboration is a *system* approach. It takes *all* model uncertainties—those of training experiments, of tabulated physicochemical "constants", and of theoretically evaluated entities—and rigorously "transforms" this collective information into the most realistic bounds on model prediction(s). The "transfer" of uncertainty is accomplished directly, in a single step, from uncertainties of training-set experiments directly to model prediction. In contrast, in the two-step process, typical of essentially all other methods, uncertainty is transferred first from training set to model parameters and then from model parameters to model prediction. The intermediate "tabulation" of model parameters of the two-step methods necessarily results in a loss of some of the information content of the system, while the one-step approach of Data Collaboration produces (and rigorously proves) the tightest possible bounds on the model prediction. Mathematical details, rigorous proofs, demonstration on realistic-size systems, and illustrative examples can be found in the literature cited above. Here we turn our attention to a new problem.

Whereas Data Collaboration can establish consistency or inconsistency of a data-and-model system [17], discriminate among alternative hypotheses [27], "propagate" uncertainties [15,16], and analyze sensitivities to uncertainty levels [18], it also opens a new alternative to the "two-step" framework.

A typical two-step procedure determines the best-fit model parameters by fitting model predictions against respective experimental targets, constrained to the uncertainties of model parameters but not those of experimental targets. In other words, the uncertainties of experimental targets are used only to determine their statistical weights and not to serve as optimization constraints [3,4]. As a result, some of the model predictions fall outside the uncertainty range of experimental measurements. In these optimizations, constraints on model parameters are simple, usually just a hypercube [3]. In reality, the region of feasible solutions (i.e., those parameters values which are consistent with all the model predictions, experimental targets, and accompanying uncertainties) is much smaller than the hypercube and possibly irregular [16].

In this study, we introduce a new approach to optimizing combustion models by constraining the optimization not only to parameter uncertainties but also to the uncertainties in experimental data. Doing so is equivalent to constraining model parameters to the feasible region. Adding experimental uncertainties as constraints ensures that all model predictions fall within the respective bounds. In other words, the model is now optimized on a feasible set and the best-fit solution produces a model whose predictions deviate the least from the experimental targets while remaining within their experimental uncertainties. At the same time the results of optimization allow one to identify experimental targets that are most difficult to match as well as model parameter values that are likely to be questionable via simultaneously computed sensitivity information.

The optimization-on-a-feasible-set paradigm utilizes the built-in features of the Data Collaboration methodology, designed to accommodate a myriad of constraints [16]. Pursuing these capabilities further, we demonstrate that this framework is even capable of addressing what appear to be heuristic questions. For instance, one of the objectives in producing GRI-Mech releases [5] was to lessen the number of model parameters adjusted and to minimize the deviation of the new values from the respective literature recommendations. This was previously approached by a tedious manual procedure. Here we show that the same goal can be met by constructing an appropriate Data Collaboration objective.

2. Computational methodology

2.1. Data Collaboration

Data Collaboration is a framework designed to make inferences from experimental observations in the context of an underlying model (see citations listed above). An observable, Y, is experimentally measured (here a training-set target) and modeled using a chemical-kinetics system. M is a model predicting Y, and d is the measured value. We assume the measurement has an uncertainty bounded by l from below and by u from above, $l \leqslant M - d \leqslant u$. The triple of measurement d, uncertainty l and u, and model M for a given set of conditions is referred to as a dataset unit. A collection of dataset units, whose elements are indexed by e ranging from 1 to m, is referred to as a dataset.

Each model, M_e, has a functional dependence on a set of model parameters, called active variables $\mathbf{X}_e$. Different models may have different active variable sets with some in common and some unique to each. Their union is denoted as $\mathbf{X} = \bigcup_{e=1}^{m} \mathbf{X}_e$. The value x_i of each variable X_i is bounded by prior knowledge, expert-assessed uncertainties of the form $x_{i,\min} \leqslant x_i \leqslant x_{i,\max}$. Collectively, all dataset parameter values, $\mathbf{x}$, span the prior-knowledge "hypercube" $H = \{X \in \mathscr{R}^n : x_{i,\min} \leq x_i \leq x_{i,\max}, i = 1, 2, \ldots, n\}$. The component-wise projection of $\mathbf{x}$ onto the active variables $\mathbf{X}_e$ is denoted as $\mathbf{x}_e$. The subset of the hypercube satisfying $l_e \leqslant M_e(\mathbf{x}_e) - d_e \leqslant u_e$ for all dataset units is referred to as the feasible set, F. The dataset is said to be *consistent* if F is nonempty. In-depth discussion of the consistency measure can be found in Ref. [17].

In the present study, the models $M_e(\mathbf{x}_e)$ are surrogate models representing the relation between

their active parameter values $\mathbf{x}_e$ and the model predictions. The normalized forms of active parameters are defined as

$$x_i = \frac{\log A_i - (\log A_{i,\max} + \log A_{i,\min})/2}{(\log A_{i,\max} - \log A_{i,\min})/2} \quad (1)$$

for the pre-exponential factor A_i of rate coefficients k_i, and

$$x_i = \frac{h_i - (h_{i,\max} + h_{i,\min})/2}{(h_{i,\max} - h_{i,\min})/2} \quad (2)$$

for enthalpy of formation h_i. Normalized in this way, the prior-knowledge space of active parameters is an n-dimensional hypercube $H = \{X \in \mathscr{R}^n : -1 \leqslant x_i \leqslant 1, i = 1, 2, \ldots, n\}$.

2.2. Optimization methods

The optimization approach used in developing GRI-Mech releases was to minimize the weighted-least-squares objective function [4,5]

$$\Phi_{\mathrm{LS}}(\mathbf{x}) = \sum_{e=1}^{m} [w_e(M_e(\mathbf{x}_e) - d_e)]^2, \quad (3)$$

constrained only by $\mathbf{x} \in H$. The w_e in Eq. (3) is the statistical weight of target e. We will refer to this approach as method LS-H.

Method LS-H is a least-squares minimization constrained to the prior-knowledge hypercube H. The results of Data Collaboration persistently show that the feasible region of the parameter space is irregular and much smaller than the entire hypercube H. In our new approach, therefore, we constrain minimization of the same objective to the feasible set, i.e., we ask for the minimum of Φ_{LS}, subject to the following constraints

$$\begin{aligned} &-1 \leqslant x_i \leqslant 1, \quad i = 1, 2, \ldots, n; \\ &l_e \leqslant M_e(\mathbf{x}_e) - d_e \leqslant u_e, \quad e = 1, 2, \ldots, m, \end{aligned} \quad (4)$$

written concisely as $\mathbf{x} \in F$. We will refer to this method as LS-F. A prerequisite for performing the optimization is that the dataset is consistent. The methods described in Ref. [17] were employed to check the consistency of our dataset.

To attain the heuristic goal of getting a model with the least number of model parameters adjusted and with the smallest deviations of the new values from the respective literature recommendations, we introduce the following objective function

$$\Phi_{1\mathrm{N}}(\mathbf{x}) = \|\mathbf{x} - \mathbf{x}_0\|_1 = \sum_{i=1}^{n} |x_i - x_{0,i}|, \quad (5)$$

where $\mathbf{x}_0$ is a n-dimensional vector of the initial values of the model parameters and $\| \|_1$ is the one-norm. The use of the one-norm objective to minimize the number of nonzero entries in a vector is well-known and now common [28]. It offers better "optimality" for our quest as discussed in the results section. Minimizing $\Phi_{1\mathrm{N}}$ on the feasible set F is referred to hereafter as method 1N-F.

Another interesting possibility is to minimize Φ_{LS} and $\Phi_{1\mathrm{N}}$ at the same time. Since it is generally not possible to minimize both objective functions simultaneously within the problem constraints, the objectives are weighted. Given a weight ratio β, the goal is to minimize the multi-objective

$$\Phi_{\mathrm{MO}}(\mathbf{x}) = \max[\Phi_{\mathrm{LS}}(\mathbf{x}), \Phi_{1\mathrm{N}}(\mathbf{x})/\beta], \quad (6)$$

constrained by $\mathbf{x} \in F$. The weight ratio β allows us to express a measure of the relative tradeoffs between the two objective functions. We notate this as method MO-F. The primary features of all four methods are summarized in Table 1.

2.3. Sensitivity coefficients

In the process of solving the optimization problem, the methodology of Data Collaboration produces sensitivity coefficients that gauge the influence of model and training data uncertainties on bounds of model prediction [17,18]. Consider an optimization problem

$$\Phi^* = \min_{\mathbf{x}} \Phi(\mathbf{x}), \quad \text{subject to} \begin{cases} x_{i,\min} \leqslant x_i \leqslant x_{i,\max} & \text{for } i = 1, \ldots, n \\ L_e \leqslant M_e(\mathbf{x}_e) \leqslant U_e & \text{for } e = 1, \ldots, m \end{cases}, \quad (7)$$

where $\Phi(\mathbf{x})$ is an objective function (Φ_{LS}, $\Phi_{1\mathrm{N}}$, and Φ_{MO} in our study), $L_e = l_e + d_e$ and $U_e = u_e + d_e$. Sensitivity coefficients are defined as the partial derivatives of Φ^*, the minimum of an objective function, with respect to the constraining bounds $x_{i,\max}$, $x_{i,\min}$, U_e, and L_e,

$$\begin{aligned} &\nu_{x_{i,\max}} = -\frac{\partial \Phi^*}{\partial x_{i,\max}}, \quad \nu_{x_{i,\min}} = \frac{\partial \Phi^*}{\partial x_{i,\min}}, \\ &\lambda_{U_e} = -\frac{\partial \Phi^*}{\partial U_e}, \quad \lambda_{L_e} = \frac{\partial \Phi^*}{\partial L_e}. \end{aligned} \quad (8)$$

2.4. Kinetic model

We demonstrate the above four optimization methods using the GRI-Mech 3.0 dataset: 77 experimental targets and 102 active parameters (see Appendix A of Ref. [16]; a brief description of the experiment targets and optimization variables is provided in Supplementary data, Tables S1 and S2, respectively). All 77 models are quadratic functions of their active variables, and these quadratic functions were obtained in earlier work using response-surface mapping. Details on parameters, experimental targets, and models are

Table 1
Summary of optimization methods.

Method	LS-H	LS-F	1N-F	MO-F
Objective function	Φ_{LS}	Φ_{LS}	$\Phi_{1\mathrm{N}}$	Φ_{MO}
Constraints	$\mathbf{x} \in H$	$\mathbf{x} \in F$	$\mathbf{x} \in F$	$\mathbf{x} \in F$

documented at the PrIMe Data Warehouse (http://primekinetics.org). The uncertainty bounds of the 102 active parameters were kept the same as in the original work of GRI-Mech optimization, and the initial point for optimization in the present study, $\mathbf{x}_0$, was calculated from the parameters A or h of GRI-Mech 3.0 using Eqs. (1) and (2). In prior Data Collaboration analyses performed with the GRI-Mech 3.0 dataset [16–18], the experimental uncertainties were estimated in a crude, *ad hoc* manner. A more systematic effort was made in the present study to evaluate these uncertainties on an individual basis. Using this more realistic set of uncertainties, the GRI-Mech 3.0 dataset turned out to be inconsistent using a consistency check procedure [17]. Target F5, laminar-flame speed for a stoichiometric methane-air mixture at 4.9 atm [5] (Personal communication, Th. Just, 1994), was identified as one possible single source of inconsistency and hence all further calculations were performed excluding this target. In this paper, we report only the results obtained with the statistical weights w_e in Eq. (3) set to 1 for all the methods, as using different statistical weights for different targets yielded qualitatively similar results.

2.5. Programming details

The MATLAB programming language was used to implement the techniques described in this work. We used the solver *fmincon* in the Matlab Optimization Toolbox [29] for methods LS-H, LS-F, and 1N-F, and *fgoalattain* for method MO-F. On an Intel Q6600@2.4 GHz computer, it took about 10 s to run an optimization for LS-H, and 130, 310, and 130–310 s to complete an optimization including a dataset consistency check for LS-F, 1N-F, and MO-F, respectively.

3. Results and discussion

The values of Φ_{LS} and Φ_{1N} at optimal solutions of the four optimization methods are compared in Fig. 1. All optimization runs started with the same initial point $\mathbf{x}_0$. Objective function Φ_{LS} was minimized with both LS-H and LS-F methods, Φ_{1N} with method 1N-F, and Φ_{MO} with method MO-F. Among the four methods, Φ^*_{LS} obtained with LS-H was the smallest, indicating the least deviation (squared and summed over all 76 targets) of model predictions from experimental targets. However, in this case, model predictions for seven experimental targets fell outside their uncertainty bounds. Optimization on a feasible set using LS-F, 1N-F, and MO-F ensured that all model predictions fell within the uncertainties of experimental targets, though the value of Φ_{LS} at optimal solution increased. The value of Φ_{LS} at the optimal solution from method 1N-F was larger than

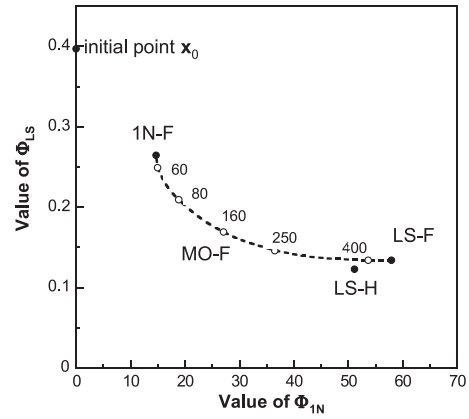

Fig. 1. The value of Φ_{LS} versus value of Φ_{1N} at optimal solutions of the four optimization methods LS-H, LS-F, 1N-F, and MO-F. The dotted line shows the values at the optimal solutions of method MO-F using different weight ratios β. Sample β values are listed along the dotted line.

those from LS-H and LS-F because Φ_{1N}, not Φ_{LS}, was minimized. The optimal value of the multi-objective optimization of method MO-F depends on the weight ratio β. As β increases, Φ_{LS} becomes dominant, and the result approaches that of method LS-F; as β is reduced toward 0, Φ_{1N} becomes dominant, and the result approaches that of method 1N-F. If β is between these two limits, a combination of Φ_{LS} and Φ_{1N} is minimized, and the values of Φ_{LS} and Φ_{1N} at the optimal solution are between those obtained from methods LS-F and 1N-F (see the dotted line in Fig. 1).

All 76 experimental targets and their corresponding optimal model predictions are shown in Fig. 2. By optimizing the model on the 76 dataset units, methods LS-F and 1N-F ensure all model predictions are within the uncertainty bounds of experimental targets. In contrast, for method LS-H, the model predictions for targets IG.1b, F6, SF7, CH3.T1b, BCH2O.T2, CO.T1b, and SCH.C13 fall outside their respective uncertainty bounds. It is pertinent to note, however, that due to the lack of information reported in experimental studies, the target uncertainties may be overestimated or underestimated. For example, the uncertainties of the laminar-flame-speed targets were estimated to be ±2 cm/s, which could be too small for SF7, 180 cm/s laminar-flame speed of a fuel-rich CO–H_2–air flame at 1 atm [30]. Sensitivity analysis showed that the upper bound of SF7 had a large influence on the optimization results of methods LS-F and 1N-F. Establishing accurate uncertainties should be the goal for the community [1]. The accomplishment we report here is the ability of optimization methods LS-F and 1N-F to find more meaningful

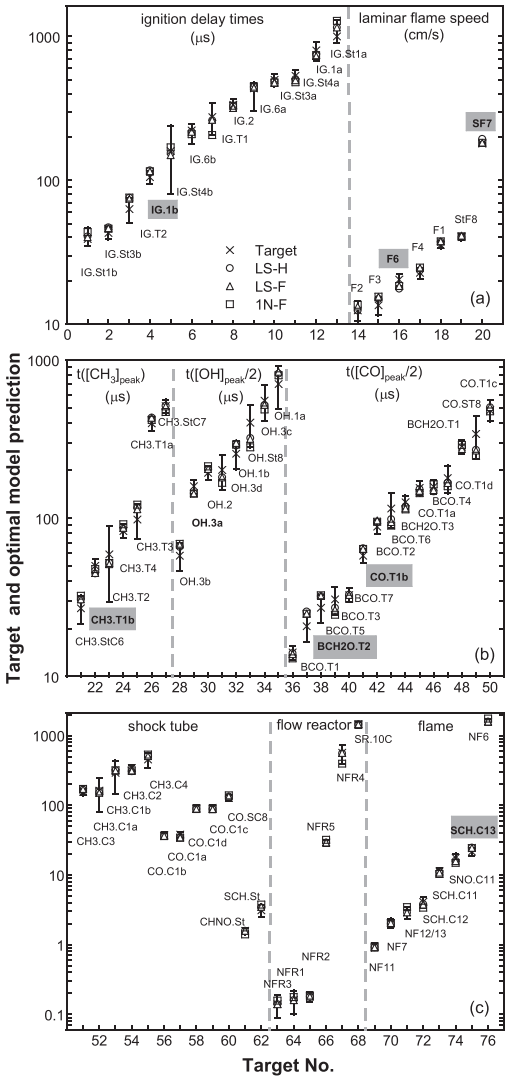

Fig. 2. Optimal model predictions using optimization methods LS-H, LS-F, and 1N-F, denoted as ○, Δ, □, respectively. × with error bars are experimental targets. Highlighted are optimal model predictions falling outside the uncertainty bounds of targets for method LS-H. Description of the targets can be found in Supplementary material.

solutions for optimal parameter values than with the "conventional" method, LS-H.

The ratios of optimal to initial values for 101 rate coefficients and one enthalpy of formation obtained with methods LS-F and 1N-F are displayed in Fig. 3. Inspection of these results indicates that the optimal values obtained with method 1N-F deviate very little from the initial values, whereas method LS-F changed many of them, often pushing them to their upper or lower bounds. Quantitatively, 62 of 102 model parameters remained unchanged in method 1N-F (within a tolerance of 10^{-6}), while only seven were left at the initial values in method LS-F. We also minimized the least-squares, $\sum_{i=1}^{n}(x_i - x_{i,0})^2$, i.e., the two-norm instead of method 1N-F's one-norm, but only 27 model parameters stayed unchanged. It thus appears that method 1N-F offers better "optimality" for our quest of finding a feasible set of parameter values that have the fewest deviations from their initial values. Whether this conclusion can be generalized should await further studies. What is important is that the framework of Data Collaboration can accommodate such seemingly heuristic questions.

We turn now to the analysis of the computed sensitivities. The optimization using method LS-H was not constrained by experimental data, and its top-ranked sensitivity coefficients with respect to parameter uncertainties were similar to those of method LS-F (they are reported in Fig. S1 of Supplementary data). Method MO-F is a mixture of methods LS-F and 1N-F. We thus focus our discussion on principal in this regard methods, LS-F and 1N-F.

Figs. 4 and 5 display the top-ranked sensitivity coefficients with respect to uncertainty bounds of model parameters, v_{xi}, obtained with methods LS-F and 1N-F, respectively. The minimum of Φ_{1N} in 1N-F, Φ^*_{1N}, is two orders of magnitude larger than Φ^*_{LS} in LS-F and hence the values of sensitivities from 1N-F are significantly larger than those from LS-F. Those model parameters that have measurable sensitivities in the optimization were pushed to either upper or lower bounds, because the sensitivities we consider here are those with respect to uncertainty bounds. For example, reaction 5, $HO_2 + CH_3 = OH + CH_3O$, whose optimal rate coefficient value was at the upper bound in methods LS-F and 1N-F, has the largest sensitivity to the upper bound, indicating the solutions of the optimization problems are very much affected by this constraint. It also implies that the upper bound of this reaction rate coefficient may be too restricted. An increase in the upper bound, which is 10^{-10} cm^3 s^{-1} and only one third of the Baulch et al. evaluation [31], will decrease the sensitivity. For reaction 12, $OH + HO_2 = O_2 + H_2O$, its initial rate coefficient value was at the lower bound, and the optimal value remained the same during optimizations (Fig. 3). Interestingly, the optimal solutions of every method in our study seemed to be quite sensitive to this constraint (Figs. 4 and 5). Fewer optimal parameter values were pushed to the limits in method 1N-F; in particular, only five variables reached upper bounds and had nonzero sensitivities (Fig. 5).

Top-ranked sensitivity coefficients with respect to uncertainty bounds of experimental targets, λ_e, obtained with methods LS-F and 1N-F are shown in Figs. 6 and 7. For method LS-F, the optimization was sensitive to six upper and four lower

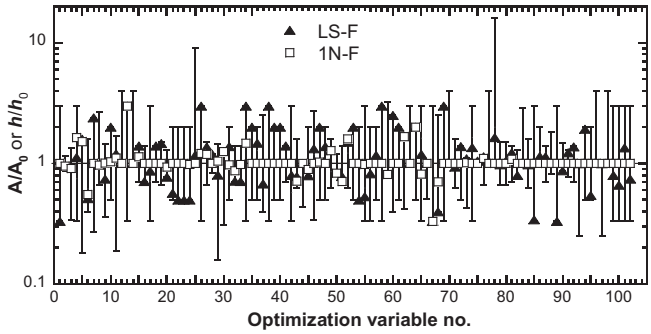

Fig. 3. Ratios of optimal to initial values for 101 pre-exponential factors of rate coefficients and one enthalpy of formation (optimization variable no. 75) obtained with methods LS-F and 1N-F. Error bars indicate the specified variable ranges. Description of the optimization variables can be found in Supplementary material.

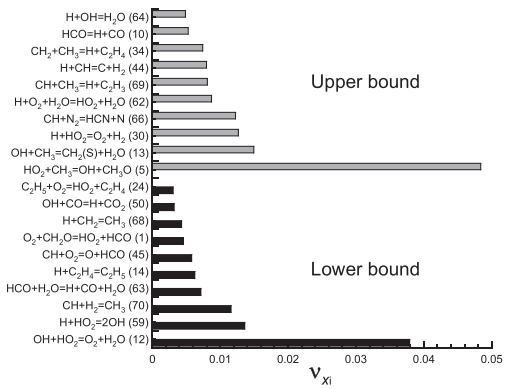

Fig. 4. Top-ranked sensitivity coefficients of Φ^*_{LS} with respect to the upper (grey bars) and lower (black bars) bounds of model parameters obtained with method LS-F. The numbers in parenthesis are the indices $i = 1, \ldots, 102$.

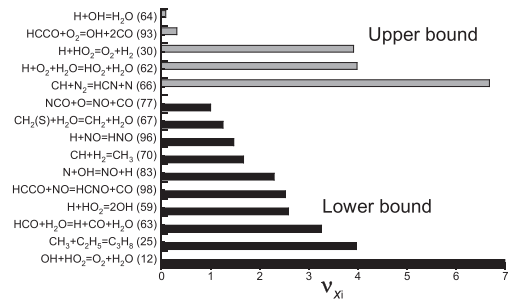

Fig. 5. Top-ranked sensitivity coefficients of Φ^*_{1N} with respect to the upper (grey bars) and lower (black bars) bounds of model parameters obtained with method 1N-F. The numbers in parenthesis are the indices $i = 1, \ldots, 102$.

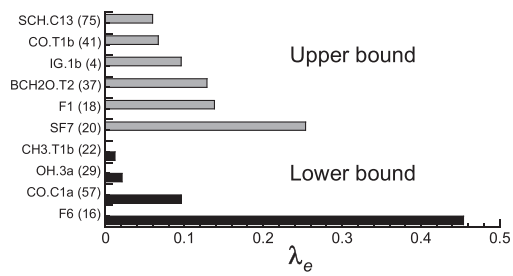

Fig. 6. Top-ranked sensitivity coefficients of Φ^*_{LS} with respect to the upper (grey bars) and lower (black bars) bounds of targets obtained with method LS-F. The numbers in parenthesis are the indices $e = 1, \ldots, 76$.

bounds (Fig. 6), and a total of 10 optimal model predictions were pushed to the uncertainty bounds of their respective experimental targets. These 10 predictions include all those that fell outside the target bounds using method LS-H (Fig. 2). There are significantly fewer constraints from targets having nonzero sensitivities than those from model parameters for method LS-F, because the model predictions are forced to match the nominal values of targets by adjusting model parameters, many of which are pushed to uncertainty limits. On the other hand, for method 1N-F, many more constraints from experimental targets show substantial sensitivity (Fig. 7), as it is aimed to minimize the deviation of optimal parameter values from initial values. This implies that more optimal model predictions are pushed to the bound limits of targets as confirmed from the results shown in Fig. 2. For both methods, LS-F and 1N-F, the lower bound of experimental target F6 shows the largest sensitivity, followed by target SF7. We suspect the actual uncertainty ranges of these two laminar-flame speeds might be larger than we estimated.

We may artificially change the bounds of model parameters or experimental targets that have the largest sensitivities to further examine the effects of these bounds. As an example, we considered experimental target F6, which has the largest sensitivity at

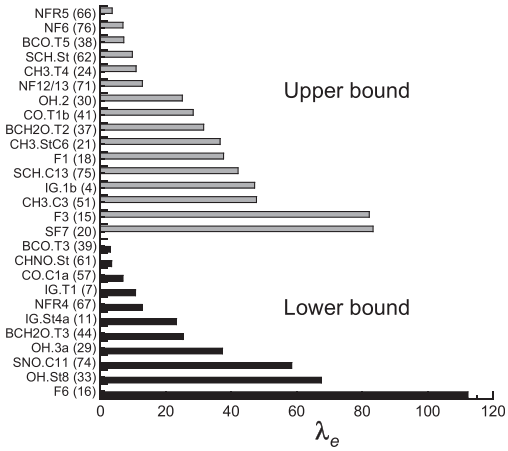

Fig. 7. Top-ranked sensitivity coefficients of Φ^*_{1N} with respect to the upper (grey bars) and lower (black bars) bounds of targets obtained with method 1N-F. The numbers in parenthesis are the indices $e = 1, \ldots, 76$.

the lower bound. We changed the uncertainty of the flame speed F6 from ± 2 cm/s to ± 4 cm/s and performed optimizations with methods LS-F and 1N-F. The results indicated a noticeable decrease in the minimum of objective function for both LS-F and 1N-F: Φ^*_{LS} decreased from 0.134 to 0.129 for method LS-F, and Φ^*_{1N} reduced from 14.7 to 12.6 for method 1N-F. This implies that the optimization on a feasible set is data-centric, in the sense that the knowledge of experimental uncertainty has direct influence on the results of optimization.

Freezing some model parameters is often used in optimization of kinetic models due to the large number of parameters. For example, GRI-Mech 3.0 was optimized using an LS-H approach, allowing only 31 model parameters to change [5]. However, LS-H optimization is likely to lead to optimal parameter values outside the feasible region. Of the methods presented here, method 1N-F found a feasible set of parameter values with the least number of varied parameters from the initial unfeasible point. This methodology could be used as guidance for choosing which variables to freeze, while keeping the optimization within a feasible set.

4. Summary

We demonstrated here new approaches to combustion chemistry model optimization—those constrained to the feasible set. Typically, model optimization is performed by fitting model predictions to pertinent training data and constrained to the prior knowledge (i.e., uncertainties) of model parameters. Optimization on the feasible set introduces additional constraints, uncertainty bounds of the training data, limiting the possible optimal solutions to the more physically relevant feasible set.

Embedded into the framework of Data Collaboration, the new optimization method allows one to examine the influence of parameter and experiment uncertainties on the optimal solution and to identify experimental targets that are most difficult to match as well as model parameter values that are likely to be questionable. The proposed methodology is flexible enough to formulate and answer questions as rich as "which minimal set of parameters needs to be adjusted to bring the model into consistency with all observed training data".

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.016.

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